



INTRODUCTION TO LINEAR ALGEBRA.

29 May 2026

Linear Algebra

- **Definition:** Linear algebra is the branch of mathematics that deals with vectors, vector spaces, linear transformations, and systems of linear equations.
- **Purpose:** It provides a framework to understand the properties and operations of these mathematical objects, which can be represented using matrices and vectors.
- **Importance:** It is a **foundational concept for Machine Learning (ML), Deep Learning (DL), Natural Language Processing (NLP), and Computer Vision (CV)**. Key concepts include scalars, vectors, matrices, mathematical operations on matrices, Eigenvalues, and Eigenvectors.

Applications

1. Data Representation and Manipulation

- **Datasets:** When creating a model, data is stored in vectors. Vectors become multidimensional as more features (parameters) are added to a dataset, with each feature representing one dimension of the vector.
- **House Price Predictor Example:** Features like Area, Rooms, and Location are **independent or input features**, while the Price is the **output or dependent feature**.

Independent or Input Features			Output or Dependent Feature
Area	Rooms	Location	Price
[1000	4	Mumbai]	-----> [1.5cr]

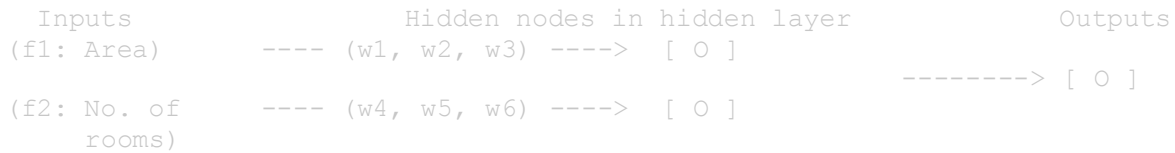
- These mathematical values—and even text (like "Mumbai")—are converted to numbers and stored in vectors.
- **Quantifying Relationships:** Mathematical operations quantify the relationship between parameters to give us the output. **Quantifying this relationship means measuring or expressing how strongly two or more variables are related using numbers** (which involves correlation, a part of statistics).
- **Model Logic:** These relations are built so the model can "think" about which parameter will affect the output and by how much. Just as human brains are trained to know that increasing a certain parameter will affect the output, the model is trained to do the same mathematically.
- Ultimately, linear algebra represents data in vectors and matrices and manipulates them through mathematical operations to analyse and extract insights.

2. Machine Learning and Artificial Intelligence

- **Model Training:** Relies heavily on linear algebra, specifically through matrix multiplication and matrix arithmetic operations.
- **Dimensionality Reduction:** Linear algebra works well with higher-dimensional data. Techniques like **Principal Component Analysis (PCA)** use **Eigenvalues and Eigenvectors** to

reduce data from a higher dimension (e.g., 500 dimensions/features) to a lower dimension (e.g., 2 or 3 dimensions) so it can be more easily understood.

- **Neural Networks:** Both forward propagation and backward propagation rely on matrix multiplication. Inputs are multiplied by the weights of hidden nodes in hidden layers using matrix arithmetic.

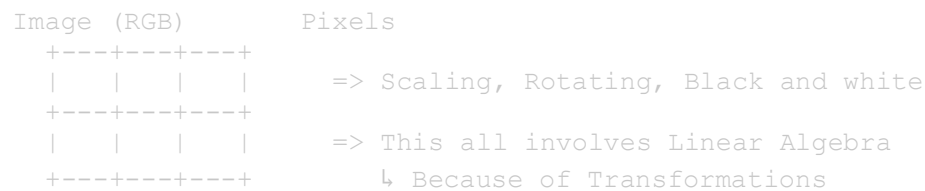


Matrix Multiplication Equivalent:

- **Hardware and Libraries:** GPUs perform faster because they have multiple cores that execute matrix multiplication in parallel. Libraries like Tensor Flow convert matrix values into tensors to process them.

3. Computer Graphics

- Images, such as RGB images, are made up of pixels.



- Modifying these images—such as scaling, rotating, and converting to black and white—all involves linear algebra because it relies on mathematical transformations.

4. Optimization

- Optimization involves solving mathematical models, such as linear equations.
- In algorithms like **Regression** (represented by the equation $y = mx + c$, where m is the slope and c is the intercept), the line of best fit on a graph (like predicting Prize vs. Size) is not drawn randomly.



Equation: $Y = mx + c$ -> Regression Algorithm

$\downarrow$ $\downarrow$
 Slope Intercept

- It is mathematically solved to find the optimal values where the maximum fit is achieved and errors are minimal.